

F642 — THE FINAL CALCULATION (pull 07-22k): done, and it resolves the fork to DIRAC. Explicit Clifford computation: the isotropy $SO(2)$ generator (= holomorphicity = the complex structure J) ANTICOMMUTES with the 4D chirality γ^5 , because the physical time direction is shared between γ^5 (spacetime volume) and the isotropy $SO(2)$ (rotation of the 2 timelike directions). Anticommuting involutions $\nexists$ a holomorphic (definite- $SO(2)$ -charge) state is a 50/50 mix of 4D chiralities $\nexists$ the holomorphic sector is a 4D DIRAC fermion $\nexists$ VECTOR-LIKE. Born=Bergman on the fermions alone does NOT give a 4D Weyl and does NOT close parity. Answer: the Weyl branch fails; parity finishes DERIVED-CONDITIONAL on the self-dual/gravi-weak input (Elie 4780 verified that route gives the chiral coupling). Geometric, non-GUT. The internal $\leftrightarrow$ spacetime alignment the arc hoped for is not just unproven — it is false at the operator level (the two chiralities anticommute = mutually unbiased). Elie's ± 4 (Dolbeault index) is real but is the DOMAIN chirality, orthogonal to spacetime γ^5 .

Lyra, Wed 2026-07-22 ~12:32. Casey: “do the final calculation.” Done. It's a clean negative for the Born=Bergman-alone hope, and a clean derived-conditional finish. Computed, not asserted.

The setup (explicit)

$SO(5,2)$: 7 directions, signature $(5,2) \rightarrow$ spacelike $\{1,2,3,4,5\}$, timelike $\{0,6\}$. - Isotropy $SO(2)$ = the compact rotation of the two timelike directions = $J_{\{06\}}$; spinor generator $\Sigma_{\{06\}} = \frac{1}{2}\Gamma_0\Gamma_6$. Its eigenvalue = the conformal weight c = holomorphicity (the complex structure J acts as this $SO(2)$). - Spacetime Lorentz $SO(3,1) = \{0,1,2,3\}$ (time 0, space 1,2,3). 4D chirality $\gamma^5 = \Gamma_0\Gamma_1\Gamma_2\Gamma_3$. The physical time direction 0 is shared: it is in the spacetime set (so γ^5 contains Γ_0) AND in the isotropy timelike 2-plane $\{0,6\}$ (so $\Sigma_{\{06\}}$ contains Γ_0). This sharing is forced: there are only 2 timelike directions, the isotropy $SO(2)$ rotates both, and Lorentz must use one of them for its time.

The computation

Clifford commutation rule: for monomials Γ_S, Γ_T (S, T index sets), $\Gamma_S\Gamma_T = (-1)^{|S||T|-|S\cap T|} \Gamma_T\Gamma_S$.

$$\Sigma_{06} = \frac{1}{2}\Gamma_0\Gamma_6, \quad S = \{0, 6\}; \quad \gamma^5 = \Gamma_0\Gamma_1\Gamma_2\Gamma_3, \quad T = \{0, 1, 2, 3\}.$$

$$|S| = 2, |T| = 4, S \cap T = \{0\} \Rightarrow |S \cap T| = 1. \quad (-1)^{|S||T|-|S\cap T|} = (-1)^{8-1} = (-1)^7 = -1.$$

$$\boxed{\{\Sigma_{06}, \gamma^5\} = 0} \quad \text{— the isotropy } SO(2) \text{ ANTICOMMUTES with 4D chirality.}$$

(Check on the boosts too: $\Sigma_{\{01\}} = \frac{1}{2}\Gamma_0\Gamma_1$, $S = \{0,1\}$, vs $\Sigma_{\{06\}}$: $|S||T|-|S\cap T| = 2 \cdot 2 - 1 = 3$ odd $\rightarrow$ boosts also anticommute with $\Sigma_{\{06\}}$ — i.e. holomorphicity is not even Lorentz-invariant, as expected of the conformal Hamiltonian.)

What it means (decisive)

$\Sigma_{\{06\}}$ and γ^5 are both (proportional to) involutions and they anticommute $\nexists$ they are mutually unbiased (like σ_x and σ_z). Consequences: - A state of definite holomorphicity (definite $\Sigma_{\{06\}}$ -charge, i.e. the Born=Bergman holomorphic sector) has $\langle \gamma^5 \rangle = 0$ — it is an equal mixture of both 4D chiralities. - The 4-dim holomorphic

sector $4_{\{c=+1/2\}}$ therefore contains 2 left-Weyl + 2 right-Weyl = a 4D DIRAC fermion. VECTOR-LIKE. - Equivalently: holomorphicity (the internal/domain chirality) is orthogonal to the spacetime chirality γ^5 . The internal $\leftrightarrow$ spacetime alignment the whole arc needed is false at the operator level — not merely unproven.

Reconciling with Elie's ± 4 (no contradiction)

Elie 4777 / the F638 chamber theorem give net Dolbeault index ± 4 — a genuine, nonzero DOMAIN chirality (the $\Lambda^{\{0,q\}}$ grading on the 10-real-dim D_{IV}^5 , i.e. the $SO(10)$ -type chirality of the tangent). That is real and correct. But it is NOT the 4D spacetime chirality γ^5 — and this computation shows the two are mutually unbiased (anticommuting). So: Born=Bergman gives a chiral *domain* spectrum, which does not translate to a chiral *spacetime* spectrum. The domain-chiral sector is spacetime-Dirac. This is the precise, computed resolution of the internal-vs-spacetime crux — and it is negative.

The verdict on the fork (F641)

- Weyl branch (Born=Bergman closes parity, no input): FAILS. Holomorphic $\nexists \gamma^5 \nexists$ the holomorphic sector is 4D-Dirac (vector-like). F640's antecedent ("Born=Bergman $\rightarrow$ 4D Weyl") is false, confirmed by computation — not merely unproven. My F640 confidence was wrong; F641's caution was right; this settles it.
- Dirac branch (self-dual/gravi-weak input needed): THIS IS THE ANSWER. 4D parity violation requires a chiral CONNECTION (the self-dual/holomorphic gauge connection), which Elie 4780 verified gives $L=0, R=2$ (chiral coupling). That input is geometric (the self-dual half of the $SO(5,2)$ connection, cross-linking BST gravity) and non-GUT — Five-Absence-safe.
- $\nexists$ Parity finishes DERIVED-CONDITIONAL on the self-dual-connection (gravi-weak) input. NOT closed on Born=Bergman alone. Honest, and it does finish — no third reframe.

The honest headline

"Why is the world left-handed?" finishes as: the fermions have a chiral DOMAIN structure (Born=Bergman, a theorem) and the SM internal content $((2,1) \oplus (1,2)$ doublet+singlet, F639), but 4D spacetime parity violation requires the weak CONNECTION to be self-dual — because holomorphicity is orthogonal to 4D chirality (this note). So parity = derived-conditional on the one geometric input (self-dual connection), exactly as gravi-weak says, now with the reason the fermion-only route can't do it *computed*. The clean "Born=Bergman alone closes it" was the hope; the calculation says no.

What stands / what this closes

- BANKED (unaffected): chiral split (n_C odd); the chamber/Dolbeault chirality mechanism as a theorem (F638, a DOMAIN statement — still correct); CP-free; custodial/ $\rho \approx 1$ /no- W_R (T2520); $1/N_c$ fractionalization (T2521); charge-row hypercharge handle.
- CLOSED (this calc): the Weyl-vs-Dirac fork $\rightarrow$ Dirac. Parity = derived-conditional on the self-dual/gravi-weak connection (geometric, non-GUT). The Born=Bergman-alone hope is refuted by computation.
- STILL OPEN (separate, unchanged): which $SU(2)$ is weak (needs the orientation on the connection); hypercharge origin (center, K806); doublet/singlet chamber sorting (F638 = flavor addresses).

Tiers / handoffs

- @Elie — verify the load-bearing anticommutation: $\{\Sigma_{\{06\}}, \gamma^5\} = 0$ with $\Sigma_{\{06\}} = \frac{1}{2}\Gamma_0\Gamma_6$ (isotropy $SO(2)$, the two timelike dirs $\{0,6\}$) and $\gamma^5 = \Gamma_0\Gamma_1\Gamma_2\Gamma_3$ (spacetime $\{0,1,2,3\}$), sharing the time direction 0. Confirm the holomorphic $4_{\{+1/2\}} = 2$ L-Weyl + 2 R-Weyl (Dirac, $\langle \gamma^5 \rangle = 0$) under $SO(3,1)$. This is the final check, and it says Dirac. Your 4780 (self-dual connection $\rightarrow$ chiral coupling) is then the operative mechanism $\rightarrow$ derived-conditional.
- @Keeper — the final calculation resolves the fork to DIRAC: the isotropy $SO(2)$ (holomorphicity) anti-commutes with 4D γ^5 (shared time direction), so the holomorphic sector is spacetime-vector-like. F640's

Born=Bergman-alone closure is refuted by computation (F641's caution vindicated). Parity finishes derived-conditional on the self-dual/gravi-weak input — geometric, non-GUT, Elie 4780 verified. This is a clean, honest finish (not a dead end): the self-dual-connection route is the answer, and we now know *why* the fermion-only route fails (holomorphic $\nexists \gamma^5$). Please close the thread on the board as: parity derived-conditional (self-dual connection); Born=Bergman-alone refuted. The chamber theorem (domain chirality) stays banked; it's a domain statement, not 4D.

- @Grace — render the finish: the two chiralities (domain/holomorphic vs 4D/ γ^5) are orthogonal (anticommute, shared time direction) — so Born=Bergman gives a chiral *domain* spectrum but a *vector-like* 4D spectrum; 4D parity needs the self-dual connection (gravi-weak, geometric). Parity = derived-conditional. Don't render "Born=Bergman closes parity" — it's refuted.
- @Casey — I did it, and the answer is Dirac: holomorphicity and 4D chirality anticommute (they share the time direction — there are only two timelike directions, the complex-structure $SO(2)$ rotates both, and Lorentz has to borrow one), so a holomorphic state is a 50/50 mix of left and right — a Dirac fermion, vector-like. So Born=Bergman *by itself* does not make the world left-handed. It gives the fermions the right internal content and a chiral *domain* structure, but 4D parity violation needs the weak *connection* to be self-dual — the gravi-weak input, which is geometric and non-GUT, and which Elie already checked delivers the chiral coupling. So the thread finishes honestly: parity is derived-conditional on one geometric input, and the pretty "closes on Born=Bergman alone" is refuted by the calculation — the two chiralities just don't line up. The discipline that kept me from banking it an hour ago was right.

Notes only; no toys/theorems claimed. Final calculation: $\{\Sigma_{06}, \gamma^5\}=0 \rightarrow$ holomorphic sector 4D-Dirac (vector-like) $\rightarrow$ Born=Bergman-alone REFUTED $\rightarrow$ parity derived-conditional on self-dual/gravi-weak connection (geometric, non-GUT). — Lyra